

Harmonizer Partwriting Guide

What the partwriting engine does today, rule by rule and weight by weight. It covers what 2.32 added (lines judged a phrase at a time), what 2.33 added (rhythm of the voices' own), what 2.34 added (repeated music keeps its parts) what 2.35 added (melody sections, chords-only passages and rewriting a range of measures) what 2.37 added (a tune in the bass, joins between sections written from both sides, a chords-only top line chosen with the voices under it) and what 2.38 added (costs and marks for contrary and hidden fifths and octaves, overlaps, cross-relations and augmented seconds, and notes spelled from their chord), how the engine measures against the September plan and against the composers in the Examples, and the defects fixed along the way.

Short version. Generate parts reads the whole piece first: local key, secondary dominants, phrase ends and cadences, non-chord melody notes and suspensions. 2.33 It also splits a held melody note wherever the chord changes under it. Then it plans the bass line on its own. It builds a wide pool of voicings for every event and keeps a shortlist that preserves each voice's melodic options 2.32. It picks the whole sequence at once with a search that scores three chords at a time.

Two passes follow. A **refine pass** 2.32 works over the draft as lines, with costs that need a whole phrase of one voice to judge: long held notes, cramped lines, repeated high points, back-and-forth, shadowing the melody, one voice doing all the moving. Its moves include re-planning a voice across a phrase and exchanging notes between voices. A **repeat step** 2.34 finds music that comes back and gives it the parts it had the first time. An **added-notes pass** 2.33 then gives the voices notes of their own where the melody holds: passing and neighbour notes, suspensions and filled leaps.

Almost every rule is a weighted cost; only a handful are absolute. Three style presets change the weights, an Added notes menu sets how busy the voices are, and a Repeated music menu sets how closely repeats follow their first statement. Everything below describes [harmonizer-2.38.html](#).

2.35 The tune can move to another stave for a passage, or there can be none (chords only). The engine itself is unchanged: Generate parts runs it once per tune stave on a view of the song in which that stave is the melody, and copies back only that stave's passages (section 3.12). **Rewrite** uses the same machinery on a range of measures and a chosen set of staves. A song with no melody sections goes through exactly the 2.34 path, note for note.

2.37 A tune in the lowest stave is now the bass line itself: no bass is planned and every other stave is an upper voice above it. After every passage is written, each join between sections is written again from both sides. A chords-only passage tries a few top lines and keeps the one whose whole texture costs least. Songs without melody sections still get exactly the 2.34 notes.

2.38 Four more partwriting faults are costed in the search, by style: fifths and octaves by contrary motion, voice overlaps, cross-relations and augmented seconds (section 3.5), and Open's parallel-fifth cost goes from 12 to 30, Close's from 4 to 5. Generated notes are spelled from their chord, not from the key signature's side alone (step 9). The same checks, with hidden fifths and octaves, can be marked on the score. The notes change in every style.

HARD never happens (filtered out, or a cost of 10^6)

STRONG a large cost; broken only when everything else is worse

SOFT a preference, weighed against the others **2.32** **2.33** **2.34** **2.35** new in that version

Costs are unitless and only their relative size matters. Pitches are sounding MIDI numbers (middle C = 60), so a tenor-clef note counts an octave below where it is written. Where a value differs by style it is given as *close* / *hymn* / *open*.

1. The pipeline

Everything lives in `generatePartsFromMelodyAndChords()` and the functions it calls;

`prepareHarmonyContext()` sets up what it shares with Refine parts. One press of Generate parts runs these steps over the whole song:

1. **Analyse** (`buildEvents`). One event per melody note, carrying the chord in force at its start. **2.33** Where the chord really changes (root or quality) during a melody note, the note becomes several events, the later ones marked *held*. The local key is inferred, secondary dominants and cadences are marked, and each melody note is classified (section 2).
2. **Plan the bass** (`planBass`). The lowest non-melody stave is the bass. **2.37** If the melody is on the lowest stave, the melody *is* the bass and this step is skipped (section 3.12). A second-order Viterbi search over every chord tone in its range picks the line (section 3.2). Nothing about the upper voices is considered yet.
3. **Build voicings** (`beamVoicingsForEvent`). For each event the staves are visited top-down with the melody and bass fixed. Each voice tries every chord tone (plus prepared 4-3 and 9-8 suspension tones) at every octave in range. Hard filters discard a candidate; costs accumulate. **2.32** Up to 48 voicings are kept per event (it was 16).
4. **Shortlist** (`shortlistVoicings`) **2.32**. The 48 are cut to at most 24 (section 3.8), keeping every pitch each voice could take, not only the best chords.
5. **Search** (`viterbiUpperVoices` → `viterbi2`). A Viterbi over pairs of consecutive choices, so a cost can depend on three events in a row. The cheapest path through the whole piece wins.
6. **Refine** (`refineVoices`) **2.32**. A local search over the finished skeleton, bass included, adding phrase-level line costs (section 3.9). It can be switched off in Settings (*Refine after Generate parts*).
7. **Repeats** (`findRepeats`, `copyRepeats`) **2.34**. Passages that come back are found; each repeat takes its first statement's voicing, the joins are checked, and a second refine pass smooths the seams while keeping the copies alike (section 3.11). Set by the song's Repeated music menu.
8. **Added notes** (`decorateVoices`) **2.33**. Passing, neighbour, suspension and chord-tone-skip notes where the melody holds (section 3.10), as many as the song's Added notes menu asks for.
9. **Write** (`materializeVoice`). Each voice is written in its own rhythm, one note per event, two where a note was added. Any length that isn't a single note value is written as tied notes. Consecutive identical pitches are tied, not re-struck. **2.38** Notes are spelled from their chord (`spellGen`, `pcSpellingFor`): a chord tone as the chord spells it (A major's third is C#, in any key), anything else as the key in force spells it (a minor key's raised 6th and 7th included), and only then by the key signature's side. Through 2.37 every note went by the key signature's side alone, so D minor's leading tone came out D $\flat$. The melody's dynamics, articulations and hairpin starts go on the first piece of each melody note; fermatas, breaths and hairpin ends go on the last.
10. **Report**. The status line counts cadences, suspensions, secondary dominants, key areas heard outside the signature, what the refine pass changed, the added notes and **2.34** the repeats kept alike. **2.32** The **Refine report** (status bar) gives each voice's line measures before and after, its

added notes, and every passage the refine pass changed, with measure numbers and reasons. Melody notes that fit no reasonable voicing produce chord suggestions (section 2.6).

Locked notes. A hand-edited note on a generated stave locks itself; notes can also be locked explicitly. Locked notes are read back as fixed pitches before the search. 2.33 An event that holds a locked note is copied back exactly as it was, rhythm and added note included, and the search works around its pitch. Hand-built Own rhythm staves are left completely alone. 2.33 Locks and the tag marking a stave as generated now survive a reload (section 8.2).

Refine parts 2.32 runs steps 6–8 on the parts as they stand. readVoiceSkeleton reads each voice back: at each event it takes the note that isn't an added note. Refine parts then chooses the added notes again and rewrites the parts. Words and marks on a part stay with any note that still starts at the same beat. A stave whose rhythm was changed by hand (a note crossing an event boundary) is heard but not rewritten.

2. What the analysis hears

2.1 Local key (inferLocalKeys)

This is Krumhansl–Schmuckler key finding, measure by measure. It uses a five-measure window weighted $\frac{1}{4} \frac{1}{2} 1 \frac{1}{2} \frac{1}{4}$ over the melody, any hand-written own-rhythm staves, and the chords (root weighted 0.9, other chord tones 0.6).

BIAS	BONUS TO CORRELATION
Key signature in force	+0.30
Relative major/minor of the signature	+0.08
Same key as the previous measure	+0.15
Minimum run for a key other than the signature	3 measures, else reverted

The local key decides which chord counts as a dominant, which note is the leading tone, and (2.33) which scale added notes are drawn from. It never changes the notation: spelling follows the written key signature.

2.2 Dominant function and tendency tones (chordTonesInfo)

A chord is a dominant when its root is 5 of the local key and its quality is major or dominant-seventh. Its third is then the leading tone. The fourth tone of any seventh chord counts as the chord seventh. A vii° or $\text{vii}^\circ 7$, and a minor v , are not treated as dominants.

2.3 Secondary dominants (markSecondaryDominants)

A chord is heard as a secondary dominant when its root falls a fifth to the next chord; it is a dominant seventh, or a major triad that is not diatonic (I, IV and V in major; III, V, VI and VII in minor); and the target is major, minor, $\text{maj}7$, $\text{m}7$ or 7 and not the tonic. The events under it are re-keyed to the target, so its third rises and its seventh falls like any $\text{V}7$.

2.4 Phrase ends and cadences (markCadences)

A note ends a phrase when it is the last note; is followed by at least half a beat of rest; is at least 2 beats long and at least $3\times$ the median note length; or ends a tonic triad that arrives on a downbeat

from V (or IV) and lasts 2 beats or more. **2.33** Length is judged on the whole melody note, even when chord changes split it into several events. An end less than 4 beats after the previous one is a breath, not a cadence. For each phrase end, the event where the final chord arrives is the cadence (2 for the piece's end); every event under that chord is *in cadence*, and the event just before is *pre-cadence*. **Picardy third:** at every cadence and on the last note, a melody singing the major third over a minor chord turns that chord major for the voicing search.

2.5 Melody notes

CLASS	HOW IT IS RECOGNISED	WHAT CHANGES
Chord tone	Pitch class is in the chord	Normal voicing; a generated voice a semitone (or major seventh) away costs <i>clash</i> .
Melody suspension	Same pitch as the previous event, across a chord change, not in the new chord, and falling 1–2 semitones to a chord tone next	Clash cost waived. Other voices avoid its resolution pitch (clash × 1.5).
Ornamental	Not in the chord, approached or left by step	Clash cost waived. Voices usually hold (holding costs nothing and the chord has not changed).
Held over a new chord 2.33	A later event of a melody note split at a chord change	Voiced like any event; if the held note isn't in the new chord it is classed as above, but never produces a chord suggestion (the writer chose that chord).
Unexplained	Not in the chord and no step either side	Voiced like a chord tone. If it lasts a beat or more, it may produce a chord suggestion.

2.6 Chord suggestions (`collectChordSuggestions`)

These are offered for an unexplained melody note of a beat or longer. The engine tries every root with every chord quality and keeps the chord that contains the melody note and shares the most tones with the written chord, merging repeats within 4 beats. They are proposed only, never applied without a click.

3. The rules as implemented

3.1 Ranges and tessitura

PART	STANDARD RANGE (SOUNDING)	PART	STANDARD RANGE (SOUNDING)
Descant	G4–E6	Tenor	C3–G4
Soprano	C4–A5	Tenor II	B $\flat$ 2–F4
Soprano II	B $\flat$ 3–G5	Baritone	G2–D4
Alto	G3–D5	Bass	E2–C4
Alto II	F3–C5	Bass II	D2–A3

RULE	STRENGTH	DETAIL
Search window	HARD	Standard range ± 2 semitones, for chord tones and added notes alike.
Extended notes	SOFT	+2 for a note in those outer 2 semitones.

Tessitura	SOFT	Comfortable zone 35–85% of the range; 0.15 per semitone below, 0.3 above. Open style moves the tenor zone down 3 and the bass down 5.
Range warnings	display	Settings → Range warnings tints notes outside the standard range (no ±2 allowance).

3.2 The bass line (planned first)

RULE	STRENGTH	DETAIL
Chord tones	HARD	Root, third, fifth or seventh in the search. 2.33 The added-notes pass may give the bass a passing note or a chord-tone skip (not a neighbour or suspension).
Position	SOFT	Root 0, third 1.5, seventh 4, fifth 6; the cadential 6/4 pays only 1.
Root at cadences	STRONG	Non-root bass: +4 on a cadence chord, +1.5 on a dominant just before one, +6 on the final chord.
Leading tone in bass	SOFT	+3 for taking it; resolving up a semitone to the tonic earns the bonus, else pays <i>unresolved</i> .
Bass motion	SOFT	Held 0.4 · step 1 · 4th/5th 1.5 · octave 1.5 · 3rd/6th 3 · tritone 6 · 7th 8 · beyond an octave 8 + 1 per semitone.
Against the melody	mixed	Parallel octaves HARD ; fifths and hidden intervals by style (3.5). Contrary motion earns 0.5 / 1.2 / 0.8.
Three-note shape	SOFT	Leap recovery (3.4) at half weight, octave leaps exempt.
Below the melody	HARD	With the melody on a higher stave, a bass above it costs 400.
Room for the voices	STRONG	Melody to bass should span at least 6 + 3 semitones per voice between; 2.5 per semitone short.
In the refine pass	SOFT 2.32	All of the above count × 2.5 there, so the refine pass trades against the bass line only reluctantly.

3.3 Building each voicing

RULE	STRENGTH	DETAIL
Chord tones	HARD	Every skeleton note is a chord tone, a prepared suspension (3.7) or a locked note. Added notes (3.10) come later.
No crossing	HARD	In the search, a voice may not sit above the voice above it (a unison costs <i>unison</i>), nor below the bass; voices above an inner-voice melody stay above it. 2.32 Only the refine pass may cross two inner voices, by a major third at most, at a cost of 4 / 7 / 5 plus 1 per semitone (a voice exchange), and never for more than two notes running (+8 each after).
Adjacent upper voices	HARD	At most 12 / 12 / 16 semitones apart.
Upper spacing	SOFT	Close: +0.4 per semitone beyond a major sixth. Hymn: none. Open: +2 per semitone under a fifth, +0.5 per semitone beyond a ninth.
Gap above the bass	SOFT	2 per semitone beyond a twelfth.
Low intervals	SOFT	4th under C3 +1.5 · M3 under G2 +5 · m3 under C3 +5 · 2nd under G3 +6, between every adjacent pair including the bass.
Unisons	SOFT	+3 / +2.5 / +5 for each voice already on that pitch.

Clash with melody	SOFT	+4 / +8 / +5 for a minor second or major seventh against a non-ornamental melody note.
Doubled leading tone or seventh	HARD	The second appearance is filtered out; the first is fine.
Complete chords	SOFT	Coverage bonus −2.5 / −4 / −3 for supplying a missing factor, × 1.6 at cadences.
Missing third	STRONG	+4, +10 on a cadence chord, +20 on the final chord. Sus chords exempt.

Doubling costs (`doublingCost`) are charged only to a voice that doubles a factor already present; a factor's first appearance is free.

CHORD / BASS	ROOT	THIRD	FIFTH	SEVENTH / OTHER
Triad, root in bass	0	2 / 5 / 3	1	3
Triad, third in bass	0	2 (min/dim) · 4	0.5	3
Triad, fifth in bass	6	6	0	6
Any seventh chord	0	5	1	3

3.4 Lines and motion (between events)

RULE	STRENGTH	DETAIL
Inner-voice motion	SOFT	Held 0 · 2nd 1 · 3rd 2 · 4th 4 · tritone 5 · 5th 3 · 6th 6 · 7th 9 · octave 3 · beyond 10 + 1 per semitone. × 1 / 1.35 / 1.
Leap recovery	SOFT	After a leap of a 4th or more: continuing the same way 1.5 (step) or 3.5 (leap); stepping back −0.8. Two same-way leaps passing an octave +3; outlining a tritone +1.2. × 1 / 1.4 / 1.
Lockstep planing	SOFT	All upper voices moving by the same interval twice running: 0 / 2.5 / 1.2.
Phrase-level line costs	SOFT 2.32	In the refine pass: see 3.9.
Forbidden melodic intervals	none	Augmented fourths and sevenths are only expensive. 2.38 An augmented second (by spelling, section 3.5) adds 3 / 12 / 8 to the minor third's cost, in the upper voices and the bass.

3.5 Parallels and hidden intervals

ERROR	STRENGTH	DETAIL
Parallel octaves and unisons	HARD	Every style, compound intervals included. A 10 ⁶ cost in the search, so locked notes can still force one. 2.33 Added notes may never form one, into the next note or at the split.
Parallel fifths	5 · 60 · 30	Close tolerates, Hymn effectively forbids, Open discourages. 2.33 Added notes may not form one in Hymn or Open. 2.38 Close 4 → 5 and Open 12 → 30: Open still wrote five across the examples, now none; Close's 5 keeps its count below 2.37's with the new costs competing.
Hidden 5ths/8ves	SOFT / STRONG	Melody-bass only. Upper voice steps: 0 / 1.5 / 1; leaps: 7 / 14 / 9. Not checked for added notes.

Contrary 5ths/8ves 2.38	2 · 12 · 8	Any pair (<code>pairTransitionCost</code>): the same perfect fifth or octave (compound included) before and after, the voices moving opposite ways, neither interval a unison and the voices not crossing. Counted in the bass plan against the melody too.
Voice overlap 2.38	1.5 · 6 · 3	(<code>checksTransitionCost</code>) For every pair with one voice above the other, a voice that moves past the note the other has just left without crossing it now: the lower rising above the upper's previous note, or the upper dropping below the lower's. The melody-bass pair is fixed and not counted.
Cross-relation 2.38	2 · 6 · 4	For each pair of pitch classes, one in the chord (or melody) before and the other its chromatic alteration in the chord (or melody) now, spelled on the same letter (cached per pair of events, <code>xrelPairs</code>): paid when one voice leaves the first and a different voice arrives on the second, unless some voice sings the chromatic step itself.
Augmented second 2.38	3 · 12 · 8	A move of three semitones that is one staff step (<code>staffStepIn</code>) in the two chords' spelling: F to G $\sharp$ in A minor, B $\flat$ to C $\sharp$ from G minor to A major. Upper voices in the search, the bass in its own plan.

2.38 Marking them on the score (`findPartwritingIssues` , Settings > *Mark partwriting problems*) is separate from the costs and judges the written notes, spelling and all, at every moment where any stave starts a note: parallel fifths/octaves as before (`findParallels`), contrary ones, hidden fifths/octaves between the top and bottom staves by voice part (the top voice leaping), overlaps between staves next to each other by voice part, cross-relations between any two staves one after the other (not at once, and not when a voice makes the chromatic step), and augmented seconds within a stave (not across a rest). Divisi lines count as part of their stave. Because the marks go by spelling, an imported MIDI file spelled with sharps where flats were meant can show cross-relations and augmented seconds that aren't really there.

3.6 Tendency tones

TONE	WANTED	BONUS / COST	NOTES
Leading tone of a dominant resolving to its tonic	Up a semitone	−3 / −4 / −3, else +5 / +10 / +6	Half bonus for falling to $\hat{5}$ (inner voices). × 1.6 at cadences.
Chord seventh	Down by step	0.6 × bonus, else 0.7 × cost	No cost if the next chord still contains it.
Secondary leading tones	As above	as above	Via the re-keying in 2.3.

3.7 Suspensions in the search (v2.18)

Prepared 4–3 and 9–8 over triads, where the chord changes on a melody note and lasts into the next event: held from the event before, resolved at the event after (breaking either costs 60). Bonus −0.8 / −0.3 / −0.5, plus −2.2 on or just before a cadence. **2.33** The added-notes pass (3.10) writes a second kind that resolves *within* the event, on the beat's second half, which is where most written suspensions now come from.

3.8 The shortlist **2.32**

Before 2.32 each event kept the 16 voicings that were cheapest *as chords*, so a voicing that connected well but spaced a little worse was gone before voice leading was considered. Now the beam keeps 48, and `shortlistVoicings` cuts them to at most 24 (18 for pieces over 500 events) in this order:

1. the 8 cheapest as chords;
2. the 8 cheapest as a continuation: a forward estimate adds each voicing's cost to its best connection from the previous event's shortlist;
3. for every voice, the best voicing that puts it on each pitch it could take, taken round-robin across voices;
4. then by the forward estimate until full.

The cheapest chord is always kept, so the shortlist can only add options, never lose the old best.

3.9 The refine pass 2.32

`refineVoices` is a deterministic local search over the skeleton: try one change, re-score everything it touches, keep it if the total falls by more than 0.05. It repeats for up to 6 passes or 40,000 tries. It keeps every cost above, scored identically: `beamVoicingsForEvent` in *forced* mode scores one given voicing. It adds costs judged over a **span**, a phrase with short phrases joined to the next until the span holds six chord changes or more:

LINE COST	CLOSE / HYMN / OPEN	DETAIL
Long hold	3 / 1.8 / 3	Per chord change a voice sits through on one pitch beyond the first 3 / 4 / 3.
Cramped or sprawling line	× 3 / 2.4 / 3	Over a span of 4+ chord changes: span ≤ a major 2nd +4, ≤ a major 3rd +1.5; beyond a tenth +0.5 per semitone (upper voices).
High point	1.5 / 1.2 / 1.5	Per extra separate arrival at the span's top note (upper voices).
Back-and-forth	1.8 / 1.5 / 1.8	Each move straight back to the note before (x-y-x) after the first in a span.
Shadowing the melody	1.5 / 2.1 / 2.4	Each move beyond two in a row that follows the melody the same way by a similar step, a 3rd or 6th from it.
Sharing the motion	0.9 / 0.75 / 0.9	Per move by which the busiest upper voice outmoves the stillest in a span, beyond 2.
Crossing	4 / 7 / 5; +8	Per crossed note (see 3.3); +8 per note once crossed more than two notes running.

The bass takes the first, second, fourth and fifth rows at 0.4 weight. Weights were set on the example songs: at these values the line faults drop by most of the way, while the chord and voice-leading costs stay within a few percent of the unrefined draft.

Moves, tried in this order at each event and voice:

1. **Re-plan a voice** over up to 14 events of a span, everything else held. A small first-order search over its chord tones, charging 1.5, then 4, for each chord change sat through and each straight step back; the winner is judged by the full cost like any move.
2. **One note** to another chord tone within a fifth (this can split a held note).
3. **A whole held note**, or its tail from here, to another chord tone that fits all of it.
4. **An octave**: the held note, or the rest of the phrase.
5. **Exchange** with another voice over 1–4 notes: each takes the other's pitch class in its own octave. This is the voice exchange, including a brief inner-voice crossing.

Locked notes never move. A suspension or any non-chord tone is frozen together with the notes before and after it.

3.10 Added notes 2.33

`decorateVoices` works on the refined skeleton. Where an event is long enough, a voice may split its note and add one of its own:

KIND	WHEN	WRITTEN	BASE SCORE
Passing	The voice moves a 3rd to its next note	note · <i>step between</i> → next	−2.0 (bass −1.6); +0.3 if chromatic
Neighbour	The voice holds into its next note (not the bass)	note · <i>step up or down</i> → same note	−0.6; −1.5 if it has already held through 2+ chord changes
Suspension	The voice steps down 1-2 semitones into a new chord that doesn't contain its old note, on a beat (not the bass; Hymn only in and around cadences)	<i>old note, tied over</i> · falls to its note	−1.0; −2.2 in or around a cadence; −0.4 more on a downbeat
Chord-tone skip	The voice leaps a 4th to an octave	note · <i>chord tone between</i> → next	−1.2 if it then lands by step, else −0.7; bass × 0.8

Where. The added note takes the second part of the event (a suspension the first). Split points are $2 \rightarrow 1+1$, $3 \rightarrow 2+1$, $1.5 \rightarrow 1+\frac{1}{2}$, $1 \rightarrow \frac{1}{2}+\frac{1}{2}$, $4 \rightarrow 2+2$, and in 6/8 a dotted half → two dotted quarters. The second part is never under an eighth. Events must be at least a half note long under *Some* (a quarter in Hymn) and a quarter under *More*. Tuplets are never split. Passing, neighbour and skip notes are not written at a phrase end or into a rest. Passing and neighbour notes come from the local key's scale (melodic minor rising, natural minor falling) or from the tones of this chord and the next; a skip is always a tone of this chord.

How many. A budget per voice: *Some* 18% of eligible events (14% in Hymn), *More* 28%, the bass at 0.6 of that. Suspensions and neighbours may each take at most a third of a voice's budget. Candidates are taken best first, and scores are adjusted as choices land:

ADJUSTMENT	VALUE
Under a melody note of 2 beats or more	−0.5
Minor 2nd / major 7th against the melody	+1.2 (Hymn +2)
Same pitch class as the melody	+0.4
Voice's share so far	$+1.2 \times (\text{its added notes} \div \text{its budget})$
Same voice has one in the event before or after	+0.7 (<i>More</i> +0.2)
Another voice has one in this event	+0.6 each (<i>More</i> +0.2)

Nothing is taken at a score of 0 or more. Every candidate must also pass these **HARD** checks against what's already chosen:

- in range (± 2);
- no unison with any voice or the melody at that moment;
- no crossing of a voice it was on the other side of, and the bass stays lowest;

- no parallel octaves (fifths too in Hymn and Open), either into the next note or with another voice moving at the same split;
- two added notes sounding together not a 2nd or 7th apart;
- one added note per voice per event;
- a suspension only if the note before it is its own preparation;
- no suspension whose resolution another voice is already singing (except a 9–8 over the root in the bass), and no 4–3 over a bass third.

Locked and frozen notes are never decorated. The same song always gets the same added notes.

Words. Added notes fall where the melody holds, so copying the melody’s words to a stave (Copy lyrics) makes them short melismas on the melody’s syllable.

2.34 In *Keep alike*, a note of a repeat that matches its first statement isn’t offered an added note of its own; after the others are chosen, it takes its first statement’s, if the note it leads to matches too and every check above still passes there.

3.11 Repeated music **2.34**

Through 2.33 every passage was voiced on its own, so a phrase that came back note for note could come back with a different alto line for no musical reason. Now the engine looks for music that repeats and keeps its parts.

Finding repeats (`findRepeats`). The melody and chords are laid on a half-beat grid (one beat for pieces over 3,000 beats). Every pair of grid points at the same distance apart is compared, for every distance from the minimum length up:

TEST AT EACH GRID POINT	RESULT
Different place in the bar or meter; different chord quality; chord roots not moved by the same interval as the rest of the stretch	no match: the stretch ends
Melody moved by the stretch’s interval, a new note starting in both or neither	match
Melody different (another note, a different rhythm, a note against a rest)	soft: forgiven, within limits

The stretch’s interval is set by its first melody note whose move agrees with the chord roots’ move (within a fifth either way). A stretch is cut at three soft points in a row (more than a beat of difference) and trimmed to start and end on matches. It counts as a repeat if it is at least three beats long (four in a 4/4 or longer bar), has at least three matching melody notes, and at most a quarter of it is soft. A pickup counts from the end of a full bar, so a phrase with its pickup is matched as a whole.

Choosing. The longest matches are taken first; each claims the grid points of its later statement not already claimed. A later statement points back to its earlier one, and chains are followed (A, A’, A” all follow A, with the intervals added). Repeats linked by hand in the report are taken before anything else; repeats switched off there are left free. Each event of a repeat is then mapped to the event of its first statement sounding at the same place, and kept only if the chord matches (same quality, root moved by the repeat’s interval) and neither or both are in a tuplet.

Copying (`copyRepeats`), after the search and the refine pass. For each run of mapped events and each voice, the first statement’s notes are moved by the repeat’s interval, and by an octave if that

keeps the whole run in range (tried in the order: none, down, up). Locked notes are never overwritten. A non-chord note (a suspension) comes along only with the notes either side of it, and if any non-chord note in or beside the copy would be left without its preparation or its step-down resolution, that voice isn't copied there. Then every copied event, and the event after the run, is checked with the search's own scoring: an impossible voicing, or a parallel octave (or fifth, per style) into it, takes the copied notes back, nearest the trouble first, unless the first statement has the same fault.

Smoothing. With *Refine after Generate parts* on, a second refine pass (at most 12,000 evaluations, 3 passes) runs with one extra cost: each note of a repeat that no longer matches its first statement costs 5 in *Keep alike* and 1.2 in *Vary lightly*. The cost is counted whichever end of the pair a move touches, so the first statement can't drift away from its repeats either. Refine parts runs the same step, so a note edited (and locked) the first time through is carried to the repeat.

REPEATED MUSIC MENU	COPY	SECOND REFINES: COST PER UNLIKE NOTE	ADDED NOTES
Keep alike (default)	yes	5	as the first statement's
Vary lightly	yes	1.2	chosen afresh
Write fresh	no	n/a	chosen afresh (the 2.33 behaviour)

Report. The Refine report's Repeated music section lists each repeat ("mm. 9-16 like mm. 1-8", "mm. 3-4 like mm. 1-2 (up a whole step)", "with small differences"), the share of the repeats' notes that match their first statement, and how many added notes were repeated. A tick box per repeat switches it off (saved as `song.recurOff`, target beat ranges); a form links a passage by hand (`song.recurLinks`: first-statement beat, repeat beat, length). Changed notes appear under What changed with the reason "the same as where this music came before".

Not found: diatonic sequences (C F G, then Dm G Am) change chord qualities; repeats shorter than three beats; a melody moved by more than a fifth. Written repeat signs need none of this: the music between them is written once.

3.12 Melody sections, chords-only passages and Rewrite 2.35

Through 2.34 the melody was one stave for the whole song (`song.notes`, the stave named by `melodyPartId`), and every other stave was either written whole by Generate parts or written whole by hand. 2.35 changes the song model in two ways and leaves the engine alone.

The model. `song.melodySections` is a list of `{beat, partId, fill}`, each running to the next, like key changes. Before the first entry, and wherever an entry names the melody stave, the tune is where it always was. `partId: null` is a chords-only passage, sung to `fill`: *chord* (one note per chord), *held* or *beat*. And "generated" is now a mark on each note (`n.gen`) rather than on the stave, so one stave can hold the tune in one passage and generated harmony in another. A note without the mark is the user's: Generate parts hears it (as if locked) and never rewrites it. A 2.34 stave marked as generated has all its notes marked on loading. 2.34 can still open a 2.35 file; it ignores the sections.

Runs. The passages are grouped by tune stave: first the melody stave's passages, then each other tune stave's, then every chords-only passage together (`sectionRunPlan`). For each group, `runSection` builds a *view* of the song, the same length and with the same measures, and runs the ordinary engine on it:

- The group’s tune stave is the view’s melody. Outside its passages the view has *gaps*: rests marked so that the analysis doesn’t count them as rests in the tune (no phrase end, no cadence because the tune moved elsewhere).
- One measure either side of each passage is included if the music there is already settled (written earlier in this run of Generate parts, or outside the range being rewritten), with every note of every stave locked. The search then joins the passage to what’s there, at both ends where it can. Nothing in that measure is written back.
- In each passage, a stave is written if it has no notes of the user’s there (Rewrite: if it’s ticked). A stave that isn’t written is heard: all its notes are locked. Where such a stave rests, the search pencils in a note for it (so the others still space themselves around a full chord) and that note is discarded.
- If the view ends before the song does, its last event is not treated as the end of the piece: no automatic final cadence, no Picardy third.
- Manual repeat links count only when both ends are in the group’s passages. Repeats are found within a group, so a verse with the tune in the tenor never copies the voicing of a verse with the tune in the soprano.

Only the passages are copied back (`sliceNotes`, `spliceNotes`): each written stave’s notes between the passage’s first and last beat, cut into tied pieces if a note crosses the edge. Every copied note is marked generated. Locked notes of the stave inside the passage are kept, as always.

Voices above the tune. With the tune in the tenor, the soprano and alto are voices ranked above the melody: as in 2.31 onward, each stays above the tune and within reach of it (`maxUpperGap` × steps between them), and the bass stays below. Whether the soprano ends up a descant or a harmony line is the search’s choice, from the usual costs.

Chords-only passages. The top stave carries a line of its own (`synthTopLine`, `planTopLineOptions`), used as the view’s melody, so the other voices are written under it exactly as under a tune. Its moments are the chord changes in the passage (*chord*, *held*) or every beat, a dotted quarter in 6/8-type meters (*beat*). A Viterbi over chord tones in the stave’s range, one semitone in from the bottom and three from the top, chooses one pitch per moment:

COST	VALUE
Same note again	0 held / 0.35 per chord / 0.25 on the beat
Step (1–2 semitones)	0.6
Third (3–4)	1.3
Fourth / tritone / fifth	2.2 / 4.5 / 3.2
Sixth to octave (8–12)	4 + 0.7 per semitone over 8
Chord factor	third 0, fifth 0.3, seventh 0.45, root 0.5 (0.15 on the first and last moment)
Tessitura	<code>tessituraCost</code> , as for every voice
Joins	the move from the top stave’s note just before the passage; half the move to its note just after

2.37 Chosen with the voices under it. Through 2.36 the single cheapest line was fixed first and the voices fitted under it. Now the Viterbi above also finds the best line starting on each other note the first moment allows, and the three cheapest different lines are kept. `runSection` writes the

whole view once with each (the ordinary engine: search, refine, repeats, added notes) and scores the result with every cost the refine pass knows (`refineVoices` ' starting total over the finished skeleton), plus $4 \times$ the line's own cost. The lowest total wins. The weight 4 was set on the examples: at 2 the choice traded the top line's smoothness away for easier inner voices and the joins got worse in Open; at 4 and 6 the results were the same. It costs up to three engine runs for the chords-only group, which is short.

The line is written one note per moment, split at barlines (tied). For *chord* and *beat* each moment is marked as a new attack, and `materializeVoice` re-strikes every voice there even on a repeated pitch; for *held* a voice that keeps its pitch ties across (the 2.33 rule for held notes).

Moving the tune (`applyMelodyIn`). Setting *Melody in* for a range: with *Bring the tune with it*, the old tune stave's notes are copied into the new one, moved by whole octaves. The octave is the one with the lowest cost of: 3 per semitone outside the new voice's range, its tessitura cost (averaged), and 2 per octave away from the usual shift (-1 octave from a women's to a men's voice, $+1$ the other way, 0 otherwise). Words and marks come too. Then the old tune stave's notes there are marked generated (for Generate parts to rewrite), and the new tune stave's generated notes there lose the mark (they are the tune now).

2.37 A tune in the bass. When the view's melody is the lowest stave (by voice rank), `prepareHarmonyContext` makes it the bass: `bassPart` is the melody stave, every other written stave is an upper voice, and the tune's pitch is entered as a locked bass note at every event, so the refine pass, the repeat step and the added-notes pass all treat it as fixed. `harmonizeCore` skips `planBass` and uses the tune as the bass sequence. Because the melody and the bass are now one voice, `transitionCost` counts each upper voice's intervals against it once (as the melody), not twice, and the bass line's own costs (`bassEventCost` , `bassTransitionCost`) are zero: the tune is what it is. Everything else follows from the ordinary rules: every voice stays above the bass, the lowest upper voice is spaced against it (`lowIntervalCost` , `bassGapCost`), doubling is judged from the note the tune gives the bass (a tune on the third of the chord: don't double the third), and each voice ranked above the melody stays within reach of it. Through 2.36 the stave above the tune was planned as a second bass line, under the soprano and alto but on top of the tune, crowding it and at times sounding below it.

2.37 The joins (`joinPasses`). Groups of passages are written one after another, so at a join between two sections the passage written first had a gap on the other side: nothing to join to, and its last event was connected straight across the gap. Once every group is written, each join is visited in order and two windows are written again with `runSection` in Rewrite mode: the last measure before the join, then the first measure after it (each within its passage, and within the range when rewriting). Everything outside the window is settled, so the measure either side of it is heard with every note locked, and the window joins the music on both sides. In a chords-only passage a window takes in whole moments, so a chord held over a barline isn't cut in two, and a chords-only passage (per chord or on the beat) ends on a new attack. The staves written are the ones Generate parts wrote there: every stave without notes of the user's (Rewrite: the ticked ones), and the top stave of a chords-only passage.

A window's rewrite is kept only if the join comes out no worse by `seamScore` , over two measures each side: the sum of `voiceMotionCost` for every move of every stave, 8 per parallel octave (and per parallel fifth in Hymn; 1.5 per fifth in Close and Open), 4 for each pair of neighbouring staves

crossed (not counting the tune), and 0.3 per semitone that two upper neighbours are more than an octave apart. Otherwise the window is put back as it was.

Rewrite is `generateWithSections` with a scope: the selected measures and the ticked staves. Everything outside the range is settled, so both edge measures are heard; the tune there and every unticked staff are heard and kept. Lock and Unlock set `locked` on every note of the ticked staves starting in the range (not on the tune).

4. Style presets and the Added notes menu

The style and the Added notes level are chosen from the menus beside Generate parts and saved with the song. Older songs open as Close harmony, Some added notes.

SETTING	CLOSE HARMONY	TRADITIONAL (HYMN)	OPEN (WIDE)
Parallel fifths	5 (2.37: 4)	60	30 (2.37: 12)
Contrary 5th/8ve · overlap · cross-relation · augmented 2nd 2.38	2 · 1.5 · 2 · 3	12 · 6 · 6 · 12	8 · 3 · 4 · 8
Hidden 5th/8ve (step / leap)	0 / 7	1.5 / 14	1 / 9
Unison	3	2.5	5
Clash with melody	4	8	5
Tendency resolved / unresolved	−3 / 5	−4 / 10	−3 / 6
Coverage bonus	−2.5	−4	−3
Third doubling (root position)	2	5	3
Max adjacent upper gap	12	12	16
Spacing curve	pull toward close	none	push toward a 5th–10th
Tessitura shift	none	none	tenor −3, bass −5
Motion weight	1	1.35	1
Contrary bass bonus	0.5	1.2	0.8
Leap recovery weight	1	1.4	1
Lockstep planing	0	2.5	1.2
Suspension bonus (search)	−0.8	−0.3	−0.5
Hold: free chord changes / cost 2.32	3 / 3	4 / 1.8	3 / 3
Range · high point · back-and-forth 2.32	3 · 1.5 · 1.8	2.4 · 1.2 · 1.5	3 · 1.5 · 1.8
Shadowing · sharing · crossing 2.32	1.5 · 0.9 · 4	2.1 · 0.75 · 7	2.4 · 0.9 · 5
Added notes under Some 2.33	half notes up, 18%	quarter notes up, 14%	half notes up, 18%
Added-note suspensions 2.33	anywhere on a beat	cadences only	anywhere on a beat
Added notes: parallel fifths allowed 2.33	yes	no	no

Added notes: **None** writes the skeleton only. **Some** is the default. **More** uses a 28% budget under quarter notes and longer, in every style. Chord vocabulary does not depend on the style: maj, m, dim, aug, 7, maj7, m7, dim7, sus2, sus4.

5. Parts, divisi, melody placement and rhythm

- **Up to 8 staves.** The lowest non-melody staff gets bass treatment; every other staff is voiced as an upper voice under the same rules.
- **Divisi.** Generate parts writes one line per staff; two-note stems come only from hand entry, Combine or import.
- **Melody on any staff.** Voices above it stay above it, the bass stays below it. [2.37](#) A melody on the lowest staff is the bass (section 3.12). [2.35](#) The tune can change staff from passage to passage, or be absent (section 3.12).
- **Eligible staves.** Empty, an empty own-rhythm staff, or a previous Generate parts result. [2.35](#) With melody sections, eligibility is per passage and per note: a staff is written in a passage where it has no notes of the user's. Changing the melody, the chords, or a staff's Part or Clef clears stale harmony (on a staff that also holds the tune somewhere, the generated notes become rests and the tune stays).
- **Rhythm** [2.33](#) . A voice has one note per event: per melody note, and per chord under a held melody note. It has two where it adds a note. Repeated pitches are tied, and lengths that aren't a single note value are tied pairs. A voice still never moves faster than the melody except for its added notes, and never adds more than one note per event.

6. Against the September plan

PLANNED	STATUS	NOW
Score whole voicings; search the phrase	DONE	Beam (48), shortlist (24) and a second-order Viterbi over the whole piece, then a refine pass with phrase-level costs.
Plan the bass first	DONE	Its own Viterbi pass; the refine pass may revise it, reluctantly.
Inversions used well	PARTIAL	First inversion preferred; 6/4 kept for the cadential 6/4. Passing and pedal 6/4 not recognised.
Ranges and tessitura	DONE	± 2 extension; no top-of-range reserve for phrase peaks (the refine pass asks for one high point per span).
Spacing and low-interval table	PARTIAL	Between every adjacent pair, bass included. No acoustic-pyramid reward.
Doubling table	DONE	Charged on real doubles only, plus a missing-third cost.
Smooth lines, leap recovery	DONE	Graded motion, three-note recovery, and (2.32) span-level line costs.
Forbidden melodic intervals	MISSING	Only graded costs by size.
Progression-aware common tones	PARTIAL	Still implicit, but long holds now cost (2.32).
Parallel 5ths/8ves, every pair	DONE	Added notes included (2.33). Contrary 5ths/8ves unchecked; octave doublings not exempt.
Hidden 5ths/8ves, outer voices	PARTIAL	Melody against bass only; not checked for added notes.
Crossing; overlap; cross-relation	PARTIAL	Crossing prevented except brief inner-voice exchanges in the refine pass. Overlap and cross-relation unchecked.
Leading-tone and seventh resolution	DONE	Stronger at cadences; secondary dominants included.
Classify melody notes	PARTIAL	Chord tone, ornamental, melody suspension, held over a new chord, unexplained.
Cadences and phrases	PARTIAL	No deceptive-cadence doubling.
Non-chord tones in generated voices	DONE	Passing, neighbour, suspension and chord-tone-skip notes (2.33), beyond the plan.
Split events at chord changes	DONE	Under a held melody note, at real changes of chord (2.33).
Local key and modulation; style profiles	DONE	Key inference and secondary dominants; Close / Hymn / Open.
Five to eight parts	PARTIAL	Voices filled; no per-count doubling, split-order or low-end rules.
Locked notes	DONE	Fixed pitches; a locked beat is kept whole (2.33); locks survive reloading (2.33).
Region regeneration	DONE	Rewrite: chosen staves in chosen measures, joined to the measures either side (2.35).
Melody moving between voices; passages without a tune	DONE	Melody sections and chords-only passages (2.35). Beyond the plan.
Linter / quality report	PARTIAL	The Refine report lists line measures, added notes, repeats and what changed where (2.32–2.34). No rule-by-rule list of every cost paid.

Repeated material voiced consistently	DONE	Exact repeats, small differences and real sequences; switch off or link by hand (2.34). Beyond the plan.
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7. Measured against the examples

`tests/partwriting-metrics.js` (2.32) takes the eight example songs and uses each song's own voices only to work out its chords (Chords from voices, one per beat). It then empties every stave but the melody and regenerates in all three styles, 24 runs in all. The generated voices are measured as lines. With `ORIGINAL=1` it measures the composers' own voices the same way. Measures sample each voice at the start of every melody note, so the first six rows don't see added notes; the last row does.

AVERAGE OVER ALL GENERATED VOICES	2.31	2.32	2.33 NONE	2.33 SOME	2.33 MORE	COMPOSERS
Chord changes held on one pitch	32%	32%	33%	34%	37%	31%
Longest such hold (chord changes)	4.4	3.7	3.9	4.0	4.3	5.8
Span per phrase (semitones)	5.0	5.2	5.4	5.3	5.3	5.7
Moves shadowing the melody	18%	17%	16%	16%	16%	17%
Back-and-forth (x-y-x) moves	24%	21%	21%	21%	20%	22%
Average run in one direction	1.50	1.55	1.55	1.55	1.55	1.73
Parallel 5ths/8ves (24 runs; all in Close and Open)	52	37	33	34	51	n/a
Long melody notes (half or more) with another voice moving under them	0%	0%	32%	61%	69%	41%

Reading it. By these pitch-only measures the composers' lines were never far from Harmonizer's. The clearest difference was the last row: through 2.32 no generated voice ever moved while the melody held, against 41% for the composers. 2.33 closes that gap, partly by following chords under held notes and partly with added notes. At *Some* it moves under long notes more often than these composers do. The chords here change on every beat, which exaggerates that; with hand-placed chords it is lower. Traditional style still writes no parallel fifths or octaves at any level.

ADDED NOTES PER 100 MELODY NOTES	SOME	MORE	MIX (SOME)
Close harmony	6	42	24 passing, 18 neighbour, 23 suspension, 9 skip
Traditional (hymn)	26	37	170 passing, 48 neighbour, 82 suspension, 27 skip

Hymn's *Some* is busier than Close's because it adds notes under quarter notes, as a chorale would, and most example melodies move in quarters.

Repeats in the examples 2.34. With chords from the composers' voices, one per beat:

EXAMPLE	REPEATS FOUND
Barber, The Coolin	mm. 33-34 like mm. 1-2; mm. 35-36 like mm. 3-4
Bruckner, Christus factus est	mm. 26-28 like mm. 22-24 (up a minor third); mm. 51-52 like mm. 36-37 (down a fourth)
Bruckner, Locus iste	mm. 30-39 like mm. 1-10 (the return); mm. 17-20 like mm. 13-16 (up a whole step); mm. 23-25 like mm. 21-23 (down a whole step)
Byrd, Ave verum corpus	mm. 44-58 like mm. 29-43 (the repeated closing section)

Copland, Help us, O Lord	mm. 32-35 like mm. 1-4, and three two-bar repeats
Morley, April is in my mistress' face	mm. 32-38 like mm. 24-30
Poulenc, O magnum mysterium	mm. 15-16 like mm. 6-7
Poulenc, Salve Regina	mm. 50-51 like mm. 42-43 (with small differences); two shorter ones

In *Keep alike*, 99% of the repeats' notes match their first statement on average over the 24 runs (the rest are at the joins); in *Vary lightly*, 96%, with added notes chosen afresh. The line measures above are unchanged to the first decimal. Parallels in Close and Open went from 34 to 36; Traditional still writes none. Generating takes about 40% longer (the second refine pass), about 1.3 s per example on average.

Melody sections and Rewrite in the examples 2.35 (`tests/sections-metrics.js`; chords from the composers' voices, one per beat). Without melody sections, 2.35 writes exactly what 2.34 wrote: all 24 runs above give the same notes. Then, for each example, the second quarter of the song was given to the tenor (tune brought across) and Generate parts run again; the first quarter was made chords only in each of the three fills; and, on the composer's own score, the soprano and bass were kept and the inner parts rewritten over the whole song.

MEASURE (8 EXAMPLES)	CLOSE	HYMN
Soprano at or above the tenor tune, sampled every half beat	100%	100%
Parallel 5ths/8ves flagged, whole song, before → after the tenor section	31 → 41	0 → 2
Chords-only passages with every stave singing on every beat (3 fills × 8)	24/24	24/24
Inner parts rewritten: parallels flagged, composers' → rewritten (the composers' soprano and bass kept)	111 → 50	111 → 35
Time, all 8 (tenor section; inner parts)	8.2 s; 8.1 s	8.3 s; 7.3 s

The two in Hymn are one pair of chords in the Barber, either side of a rest of several bars in every voice, just where the tenor passage begins: the display check counts across the rest, and the engine doesn't treat notes that far apart as consecutive. Close tolerates parallel fifths by design (section 4).

The tune in the bass, the joins, chords-only lines 2.37 (`tests/joins-metrics.js`; chords from the composers' voices, one per beat, every stave but the melody generated first). For the bass, the second quarter of each song was given to the bass (tune brought across). For the joins, the second quarter was given to the tenor and the last quarter made chords only (one note per chord): three joins per song, 24 in all. Leaps are counted for every stave that isn't the tune on either side, from its last note before the join to its first after.

MEASURE (8 EXAMPLES)	CLOSE 2.36 → 2.37	HYMN	OPEN
Bass tune: another stave sounding below it (every half beat)	5% → 0%	7% → 0%	11% → 0%
Bass tune: lowest other voice within a whole step of it	10% → 2%	13% → 3%	22% → 3%
Bass tune: chord has its third	99% → 100%	97% → 97%	98% → 99%
Bass tune: parallels in the passage	12 → 10	2 → 2	1 → 2
Joins: semitones leapt across the 24 joins	205 → 188	181 → 152	205 → 202
Joins: leaps wider than a fifth	5 → 3	3 → 2	3 → 4

Joins: parallels within a beat of a join	6 → 6	2 → 2	2 → 2
Chords-only ending: top line motion (semitones)	193 → 183	193 → 181	193 → 182
Chords-only ending: the other voices' motion	1767 → 1770	1850 → 1791	1901 → 1893
Chords-only ending: parallels	7 → 8	0 → 0	2 → 2

Of the parallels at the joins in Close, four are fifths (which Close allows), one is the tune itself handing over from soprano to tenor an octave lower, and one is in the Barber across a rest of several bars. In Hymn both are in the Barber across that rest, as in 2.35. In Open, the wide leaps are now an alto octave where the tenor passage begins in the Bruckner *Christus* and a bass sixth at a join in the Byrd, and the one in *Locus iste* is gone; the Barber and Copland ones are unchanged. Songs without melody sections: all 24 runs of section 7 give the same notes as 2.36. With sections, Generate parts takes about a quarter longer (the join windows and the extra chords-only tries).

The partwriting checks `2.38` (`tests/checks-metrics.js`): the same 24 runs as the first table, every finding of `findPartwritingIssues` counted over the whole score, melody included). “2.37” is 2.37’s notes spelled the 2.38 way, so the spelling-based counts compare like with like.

FOUND (8 EXAMPLES)	CLOSE 2.37 → 2.38	HYMN	OPEN	COMPOSERS
Parallel 5ths/8ves	31 → 25	0 → 0	5 → 0	111
5ths/8ves by contrary motion	44 → 16	35 → 0	37 → 2	20
Hidden 5ths/8ves (outer voices, top leaping)	1 → 1	0 → 0	0 → 1	26
Voice overlaps	62 → 46	69 → 28	82 → 61	509
Cross-relations	58 → 37	64 → 38	67 → 42	78
Augmented seconds in the generated voices (in the melodies: 7)	20 → 6	11 → 0	25 → 3	19 in all

The composers’ column counts their own four voices (one run); their overlaps are mostly the Barber and the Poulenc *O magnum*, whose voices cross freely. What’s left in Hymn is mostly with the melody: the soprano leaping down past the alto’s last note, or a tune that moves from F to F# while the alto had the F just before. The seven augmented seconds in the melodies are MIDI spellings (A# for Bb in the Copland). Line measures over the 24 runs, 2.37 → 2.38: holds 34% → 34%, longest hold 4.0 → 4.1, span 5.3 → 5.3, shadowing 16% → 15%, leaps per 100 moves 22.7 → 21.5, back-and-forth 20% → 22%, moving under long melody notes 62% → 60%. Generating takes about a fifth longer. With melody sections (`sections-metrics.js`, Close), parallels with the tenor section fall from 41 to 28. At the joins (`joins-metrics.js`) leaps across the 24 joins go 188 → 185 in Close and 202 → 183 in Open, but 152 → 169 in Hymn, where the new costs outweigh the join’s smoothness.

8. Defects found and fixed

DEFECT	WHAT WAS FOUND	FIX
8.1 Chords under a held note ignored (2.32 and earlier)	Melody E4 (half) G4 (whole) C5 (whole) over C, F, G, Am, C in Hymn style: the voices held C, E and G straight through the F, G and Am.	2.33: the held note becomes one event per chord (2.4); the voices now move F → G → C with the harmony.
8.2 Reload dropped locks and the generated tag (older versions)	<code>normalizeSong</code> copied only the note fields it knew, so a page reload, Open or Copy lost every lock and the tag marking a stave as generated. Afterwards Generate parts said “every other stave already has its own handwritten rhythm”.	2.33: both are kept, as is the added-note tag. A song saved by an earlier version may already have lost the tag; unticking Own rhythm on the stave hands it back.

8.3 Good lines pruned before the search (2.31 and earlier)	Only the 16 voicings cheapest as chords reached the search.	2.32: the three-way shortlist (3.8).
8.4 Lines never judged beyond three notes (2.31 and earlier)	An inner voice could sit on one note for 14 chord changes (a two-chord vamp).	2.32: the refine pass (3.9); in 2.32 the same vamp held at most 4, and with added notes the held stretches get neighbour notes.
8.5 Repeated music voiced afresh (2.33 and earlier)	A 16-bar tune sung twice (verse 1, verse 2): in Close style, 14 of the 73 notes of the second verse's parts differed from the first, with no change in melody or chords.	2.34: the repeat step (3.11); in Keep alike every note of the second verse matches the first, in every style, added notes included.
8.6 Generated notes misspelled (2.37 and earlier)	Every generated note was spelled by the key signature's side alone: in D minor (one flat) the leading tone came out D $\flat$, in F major a D major chord's third G $\flat$.	2.38: spelled from the chord, then the key (step 9).

The six 2.31 fixes (chords losing their third, the bass climbing into the melody, spacing against the bass, 6/4 versus 6, the leading-tone exception, inner-voice melodies) stand and are still tested; see the 2.31 edition of this guide for details. The suite: 476 logic checks pass. Of 168 browser checks, 165 pass here; the other three need the site's shared files, which this test server didn't have.

Still open

- **2.38** Contrary 5ths/8ves, overlaps, cross-relations and augmented seconds are costed in the search, the bass plan and the refine pass (which uses the search's pair costs), but not in the added-notes pass, which still checks only parallels and crossings. Hidden intervals are not checked for added notes. Octave doublings are not exempt from the parallel-octave rule.
- Passing and pedal 6/4 and the deceptive cadence are not recognised. Non-chord generated notes are spelled from the key, not from the line they move in (a chromatic passing note may take the "wrong" enharmonic).
- Added notes come one per voice per event: no two-note passing through a fourth, no bass suspensions (2-3), no anticipations or escape tones, no imitation of melody figures. They go only where the melody holds.
- No rule-by-rule list of every cost paid. **2.35** A region can now be rewritten (3.12).
- **2.37** Melody sections: the joins are rewritten a measure each side, after the fact, and kept only when they come out smoother, rather than planned together with both passages in one search. A chords-only top line is chosen from three candidates written in full, not searched jointly with the voices note by note. A tune in the bass that is a non-chord tone leaves the chord without its real bass note there (the other voices still fill the chord above it).
- Repeats: diatonic sequences aren't found (link them by hand); a repeat's last bar leading somewhere new keeps the first statement's voicing up to the join, where the seam is smoothed rather than planned.
- Parallel octaves are a 10^6 cost, not a filter, so locked notes can force one without it being reported.

9. Code map

FUNCTION	WHAT IT DOES
HARMONY_STYLES / STYLE_CLOSE, STYLE_HYMN, STYLE_OPEN	weights (section 4)
HARMONY_MOTIONS, ORN_KINDS	Added notes levels and kinds (2.33)
VOICE_RANGE_SOUNDING, ROLE_RANK	ranges and voice order
chordTonesInfo, withPicardyThird	chord factors, tendency tones
inferLocalKeys, buildEvents, markSecondaryDominants, markCadences	analysis (section 2); events split at chord changes (2.33)
prepareHarmonyContext, collectLockedAnchors	shared set-up; locked notes and locked beats (2.33)
tessituraCost, styleRange, doublingCost, lowIntervalCost, bassGapCost	per-note costs
voiceMotionCost, bassMotionCost, secondOrderMotion	line costs
pairTransitionCost, tendencyTransitionCost, transitionCost, upperPairCost, upperTripleCost	between-chord costs
planBass, bassEventCost, bassTransitionCost	bass pass
beamVoicingsForEvent, shortlistVoicings, viterbiUpperVoices, viterbi2	voicing search (forced mode for scoring one voicing)
refineVoices, refineFixedMap, lineStats	refine pass (2.32)
findRepeats, copyRepeats, applyRepeatsStep, RECUR_MODES	repeated music (2.34)
melodySectionSpans, sectionRunPlan, runSection, generateWithSections, refineWithSections	melody sections and Rewrite (2.35)
synthTopLine, planTopLine, planTopLineOptions, fillMoments	chords-only passages (2.35); several top lines tried (2.37)
joinPasses, seamScore	the joins between sections written again from both sides (2.37)
applyMelodyIn, lockRange, sliceNotes, spliceNotes	moving the tune, locking a range, cutting and joining note lists (2.35)
decorateVoices, decoSplitPoint, splitBeats	added notes (2.33); a repeat's added notes (2.34)
materializeVoice, readVoiceSkeleton	write-out and read-back
refinePartsNow, buildRefineReport, showRefineReport	Refine parts button and report
collectChordSuggestions	chord proposals
checksTransitionCost, xrelPairs, evSpell, staffStepIn	overlap, cross-relation and augmented-second costs (2.38)
spellGen, pcSpellingFor	spelling generated notes from the chord (2.38)
findParallels, findPartwritingIssues, drawParallelMarks, rangeStatus	display-only checks (Settings); every kind from 2.38
genRuns, drawGenMarks	marks over runs of generated notes (2.38)

tests/partwriting-metrics.js, sections-metrics.js, joins-metrics.js,
checks-metrics.js

the measurements in section 7

Source: harmonizer-2.38.html. Checked 30 September 2026.